

OAS-001 — Architectural Principles

Status: Foundational

Authority: Architectural

Depends on:

- OAS-000
 - OCS-000 through OCS-032
-

Purpose

This specification defines the Architectural Principles of Organodynamics.

Architectural Principles translate constitutional requirements into architectural constraints.

They do not prescribe implementation.

They constrain architecture.

Every architectural decision SHALL be justifiable through one or more Architectural Principles.

1. Constitutional Preservation

The primary responsibility of architecture is the preservation of constitutional meaning.

Architecture SHALL preserve:

Identity

Meaning

History

Relationships

Traceability

Continuity

Architecture SHALL never optimize by weakening constitutional integrity.

2. Separation of Meaning and Mechanism

Meaning SHALL remain independent of mechanism.

Architecture SHALL distinguish between:

Semantic meaning

Behavior

Execution

Storage

Presentation

Communication

Mechanisms MAY evolve independently.

Meaning SHALL remain stable.

3. Replaceability

Every architectural component SHALL be replaceable.

Replacement SHALL preserve constitutional behavior.

Architecture SHALL assume that every implementation technology is temporary.

Durability SHALL emerge from architecture.

Not technology.

4. Explicit Boundaries

Every architectural component SHALL expose explicit boundaries.

Boundaries SHALL define:

Purpose.

Responsibilities.

Dependencies.

Public interactions.

Hidden coupling SHALL be treated as architectural debt.

5. Composability

Architectural components SHALL compose through constitutional interfaces.

Composition SHALL preserve:

Local autonomy.

Global consistency.

Historical continuity.

Independent evolution.

Composition SHALL remain explainable.

6. Progressive Complexity

Architecture SHALL remain simple until constitutional necessity requires additional complexity.

Complexity SHALL emerge.

It SHALL NOT be anticipated without evidence.

Premature architectural sophistication SHALL be avoided.

7. Event Orientation

Architectural change SHALL be represented through observable Events.

Events SHALL preserve:

History.

Context.

Identity.

Relationships.

Architectural state SHALL remain reconstructable from constitutional history.

8. Local Sovereignty

Every architectural Place SHALL remain sovereign.

Sovereignty SHALL include:

Independent evolution.

Independent memory.

Independent participation.

Independent architectural composition.

Federation SHALL preserve sovereignty.

It SHALL not replace it.

9. Explainability

Architectural behavior SHALL remain explainable.

Independent architects SHALL understand:

Why a component exists.

Why boundaries exist.

Why dependencies exist.

Why evolution occurred.

Opaque architecture SHALL be considered incomplete.

10. Architectural Humility

Architecture SHALL assume future change.

Every architectural decision SHALL acknowledge:

Known limitations.

Alternative designs.

Future evolution.

Unknown future requirements.

Architecture SHALL remain intentionally unfinished.

11. Federation by Design

Independent architectural systems SHALL be capable of federation without sacrificing local constitutional integrity.

Federation SHALL emerge through shared constitutional meaning.

Not through centralized architectural control.

12. Architectural Longevity

Architecture SHALL outlive implementation.

Implementations MAY change frequently.

Architecture SHOULD evolve slowly.

Constitution SHOULD evolve rarely.

This layered stability SHALL preserve the discipline across technological generations.

13. Independent Reconstruction

Architecture SHALL enable independent architects to reconstruct equivalent architectural intent using only:

The Constitution.

The Architectural Specification.

The Constitutional Corpus.

No privileged implementation knowledge SHALL be required.

14. Architectural Verification

Architectural correctness SHALL be evaluated by preservation of constitutional behavior.

Different structures MAY be equally correct.

Equivalent constitutional behavior SHALL remain the defining criterion.

15. Architectural Principle Test

An architectural principle satisfies this specification when an independent architect can explain:

Why this principle exists.

Which constitutional requirements it preserves.

How it constrains architecture.

Which architectural failures it prevents.

How future implementations remain free despite this constraint.

Conformance

An architecture conforms to this specification when its organization preserves constitutional behavior while allowing multiple independent implementations to emerge without constitutional divergence.

Architectural Principle

Good architecture does not attempt to predict the future.

It prepares the future to evolve safely.

The purpose of architecture is therefore not to freeze design,

but to preserve meaning while allowing every generation of builders to create new implementations without rebuilding the discipline itself.

End of Specification OAS-001